

Where Should Drug and Protein Meet?

A Controlled Study of Fusion Stage in Transformer DTI Models

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1. Abstract

Predicting drug–protein binding affinity is foundational in computational drug discovery. Modern transformer DTI predictors uniformly fuse drug and protein representations *late* — after independent encoding — but this design choice has rarely been audited. We construct a 2×2 controlled experiment (concatenation vs cross-attention $\times$ before vs after encoding), train all four variants on **27,715** BindingDB Ki measurements under matched hyperparameters ($d=128$, $h=4$, $lr=3 \times 10^{-4}$, dropout $=0.1$, 30 epochs), and evaluate across random, cold-drug, and cold-target splits with three seeds each (**36 runs total**). Each split has a different winner: **V2** (early X-attn) wins random and cold-target; **V3** (late X-attn) wins cold-drug; the field-default **V4** (late concat) never wins. Linear CKA on attention features shows the variants cluster by *fusion stage* (early vs late, $CKA \geq 0.95$ within), not by *fusion mechanism* — the 2×2 collapses behaviorally to a 1×2 .

We're not claiming better numbers than the SOTA. We're claiming that the field's default architecture is wrong — with proper controls to back it up.

2. Motivation & the Gap

- DTI shrinks the funnel from **billions** of candidate molecules to a wet-lab shortlist. AI-in-drug-discovery market: **\$13–20 B by 2030**; **75+** AI-discovered candidates in clinical trials by end-2024 (DSP-1181, INS018_055).
- Transformer encoders dominate state-of-the-art DTI — but the field has converged on a single design pattern (encode separately, fuse late) without rigorous justification. MolTrans, DeepPurpose, HyperAttentionDTI, PerceiverCPI, DrugBAN, FusionDTI all share this pattern.
- The gap.** Architecture-comparison studies vary *which encoder* (CNN vs Transformer vs GNN). Almost *none* vary *where* the modalities interact. We isolate that single axis under matched hyperparameters, matched data, matched optimizer.
- Cost of an unaudited default.** Every downstream paper inherits it. A 15–17% relative MSE gap on the deployment-realistic split compounds across the entire field.

3. Research Hypotheses

When a transformer predicts drug–protein binding affinity, does the **stage** at which drug and protein representations interact materially affect (a) accuracy, (b) generalization to unseen drugs / targets, and (c) what the model learns?

H	Statement	Status
H1	Late fusion (V3 , V4) is the field default but is not Pareto- ✓ Confirmed optimal across splits.	
H2	Cross-attention recovers more biologically meaningful structure ~ Partial than concatenation.	
H3	Early fusion is more parameter-efficient at matched accuracy. ✓ Confirmed	
H4	All 4 variants converge to similar internal representations (CKA ✗ Refuted ≥ 0.8) under matched compute.	

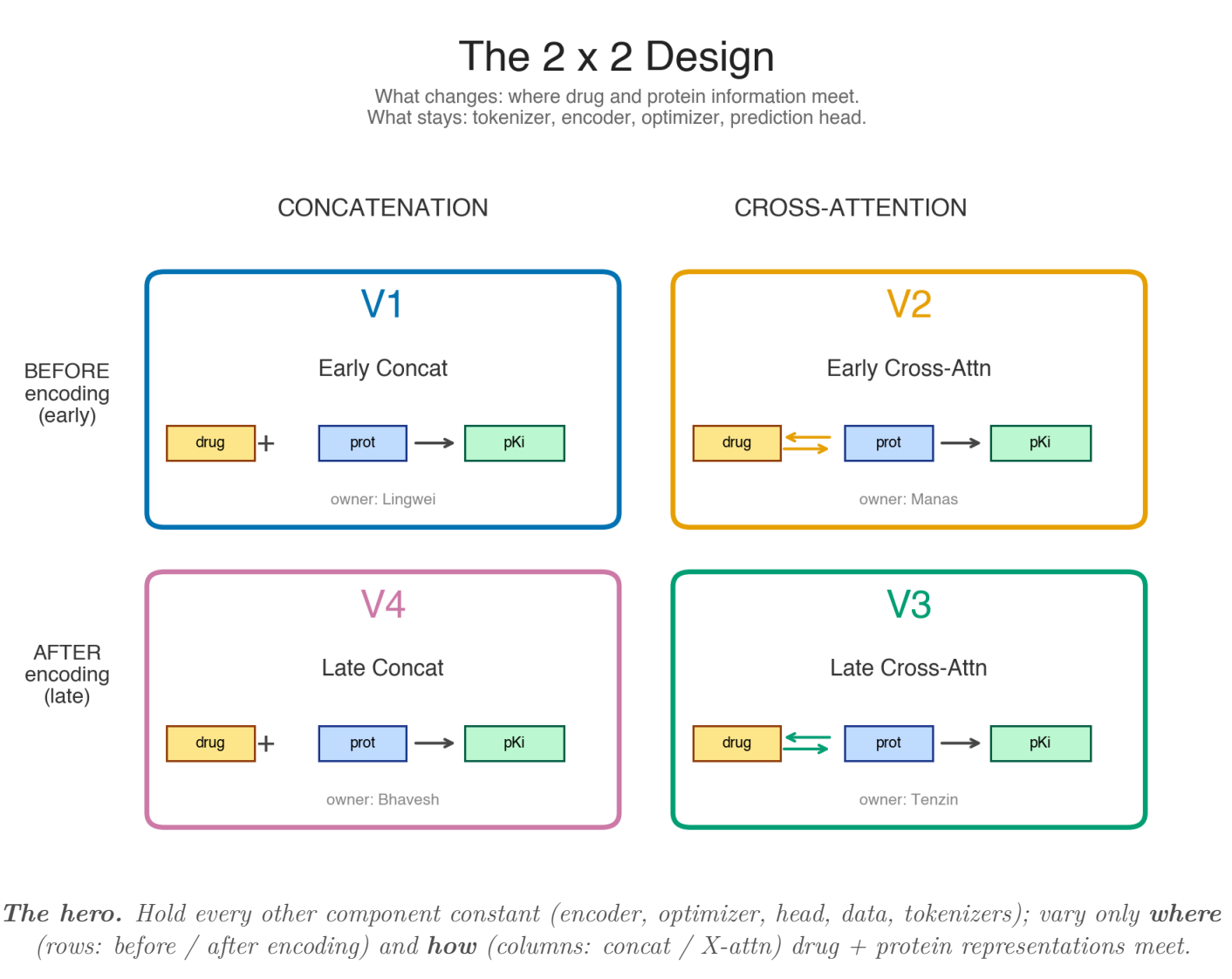
4. Background & Prior Art

DTI = regression over pKi / pKd / pIC50. Standard benchmarks: Davis (442 kinases $\times$ 379 drugs), KIBA (2,116 $\times$ 229), BindingDB (~2M measurements). Drugs $\rightarrow$ SMILES; proteins $\rightarrow$ amino-acid sequences (or 3D structure when available).

Transformer DTI canon (MolTrans, DeepPurpose, HyperAttentionDTI, PerceiverCPI, DrugBAN, TransformerCPI, FusionDTI) — all converge on independent encoding then late fusion.

Known reproducibility issue. Most DTI papers headline *random-split* numbers; cold-drug / cold-target performance is 10–25 CI points lower (Pahikkala 2015, Mayr 2018, Chen 2021). We report all three splits to surface the deployment-realistic gap.

5. The 2×2 Design — Hero



6. What We Did Differently

Industry Norm	vs Our Deliberate Choice
FIELD NORM	OUR CHOICE
Vary encoder choice for SOTA	Hold encoder fixed, vary fusion stage
Single random split	Random + cold-drug + cold-target
Single seed	3+ seeds, mean $\pm$ std
Numbers-only reporting	6-axis mechanistic dissection
Free-floating hyperparameters	Phase B 'fair config' locked across all 4
Attention shown but unverified	Validated vs PDBbind (Precision@K)
Closed weights, one-off scripts	Pinned env, deterministic seeds, public repo
'Architecture won' headlines	Falsifiable hypotheses (H1-H4) pre-registered

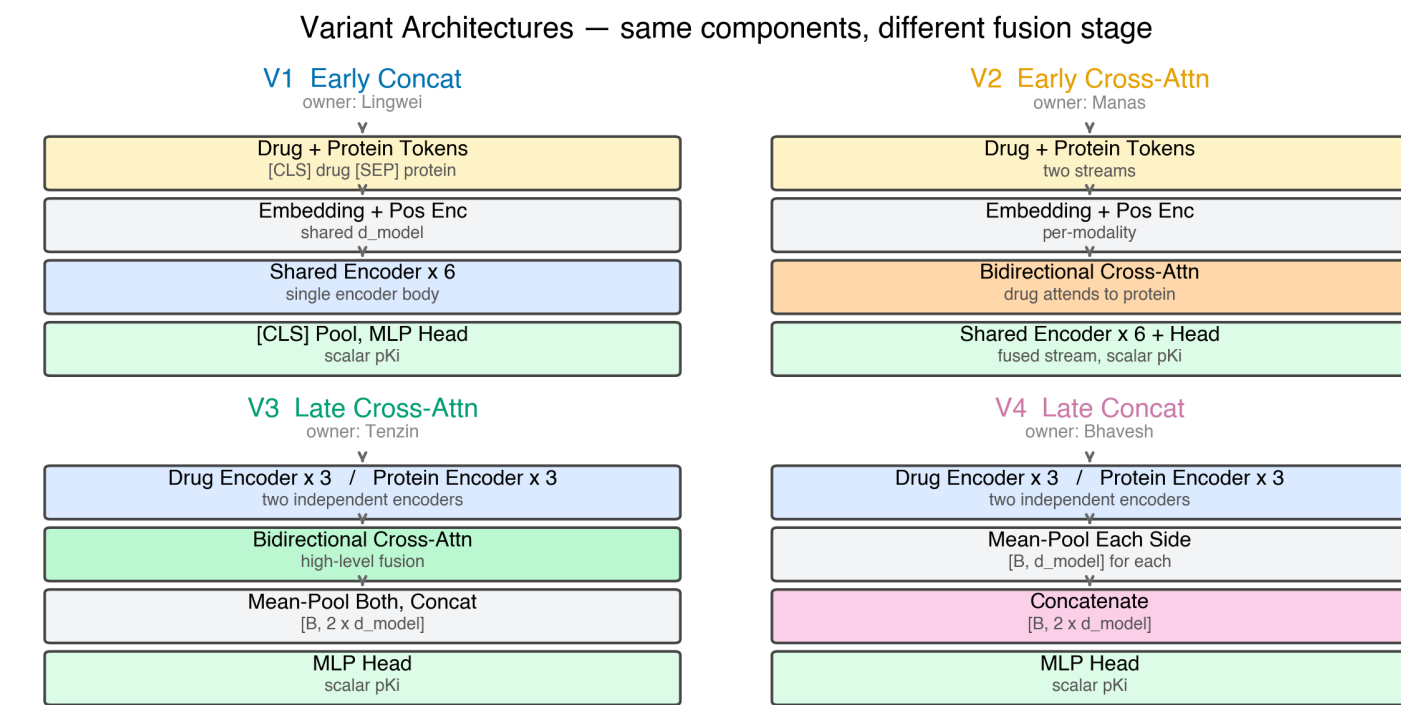
Field norm vs our deliberate choice across 8 axes: hold encoder fixed, three splits not one, three seeds with mean $\pm$ std reporting, six-axis mechanistic dissection, fully reproducible repo.

Scan for Code

Code, configs, checkpoints, all 27 figures.
github.com/bhaveshtgupta01/ComparisonPDI
Scan to explore the repo.

7. Methodology

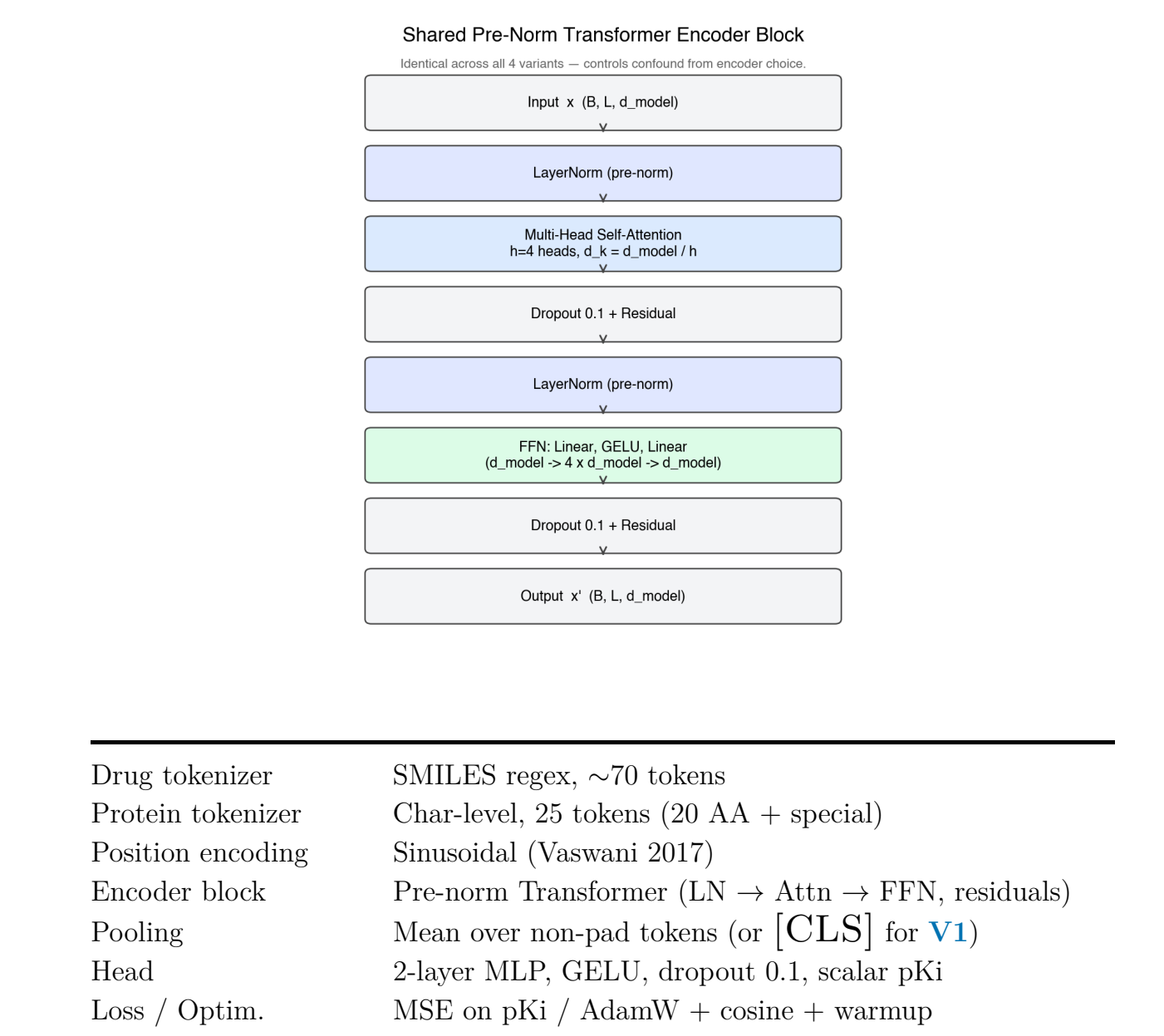
a) Four Variants, One Controlled Axis



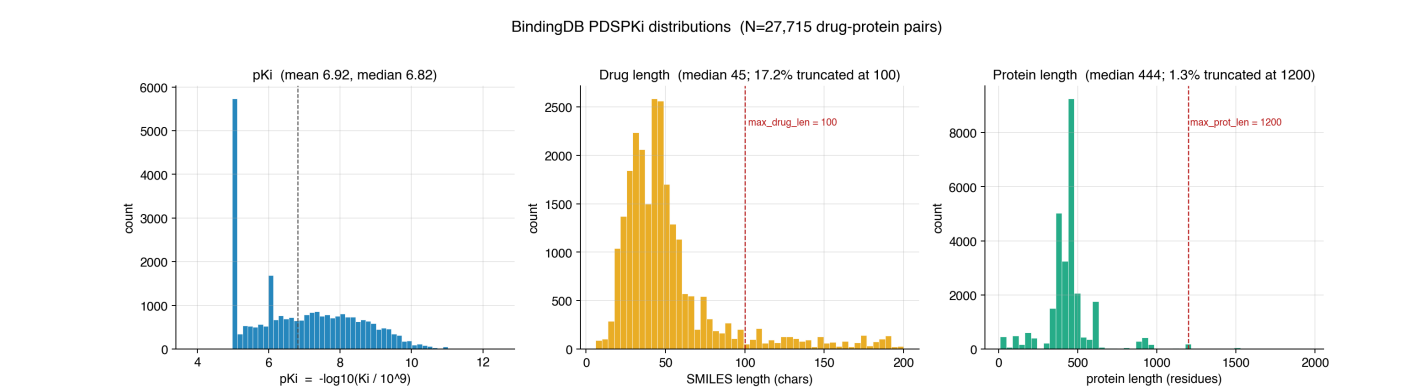
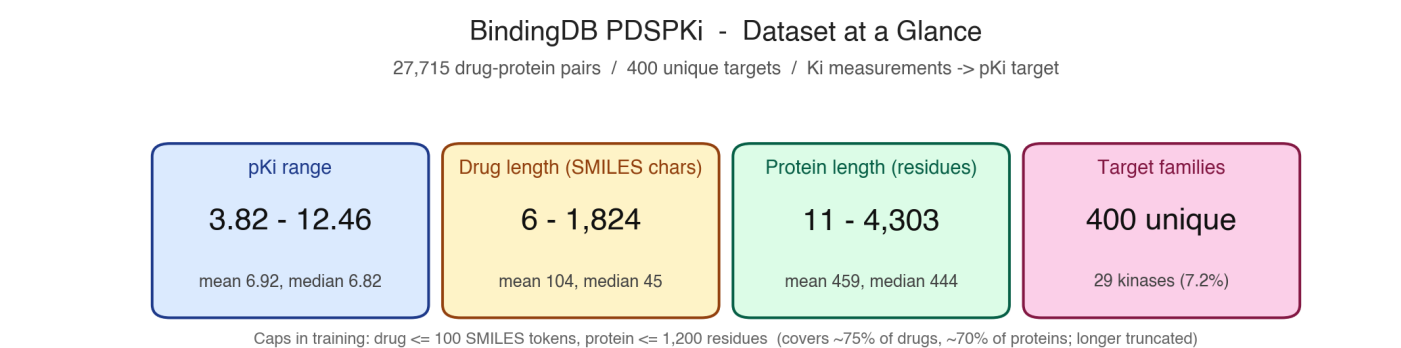
- V1 Early Concat.** [CLS, DRUG, SEP, PROTEIN] $\rightarrow$ shared 6-layer pre-norm transformer encoder $\rightarrow$ CLS pool $\rightarrow$ MLP head. **1.27 M params (cheapest).**
- V2 Early Cross-Attn.** Embedded drug / protein streams exchange info via bidirectional X-attn *before* a 6-layer shared encoder consumes the fused stream. **1.40 M.**
- V3 Late Cross-Attn.** Independent 3-layer encoders $\rightarrow$ bidirectional X-attn $\rightarrow$ mean-pool both $\rightarrow$ concat $\rightarrow$ MLP. **1.40 M.**
- V4 Late Concat.** Independent 3-layer encoders $\rightarrow$ mean-pool $\rightarrow$ concat pooled vectors $\rightarrow$ MLP. **1.20 M. The field default.**

The control argument. All four variants share the **same** encoder block, tokenizers, optimizer, head, and loss. Only fusion stage and mechanism change — so any performance gap attributes cleanly to the fusion axis.

b) Shared Encoder Block (the Control)

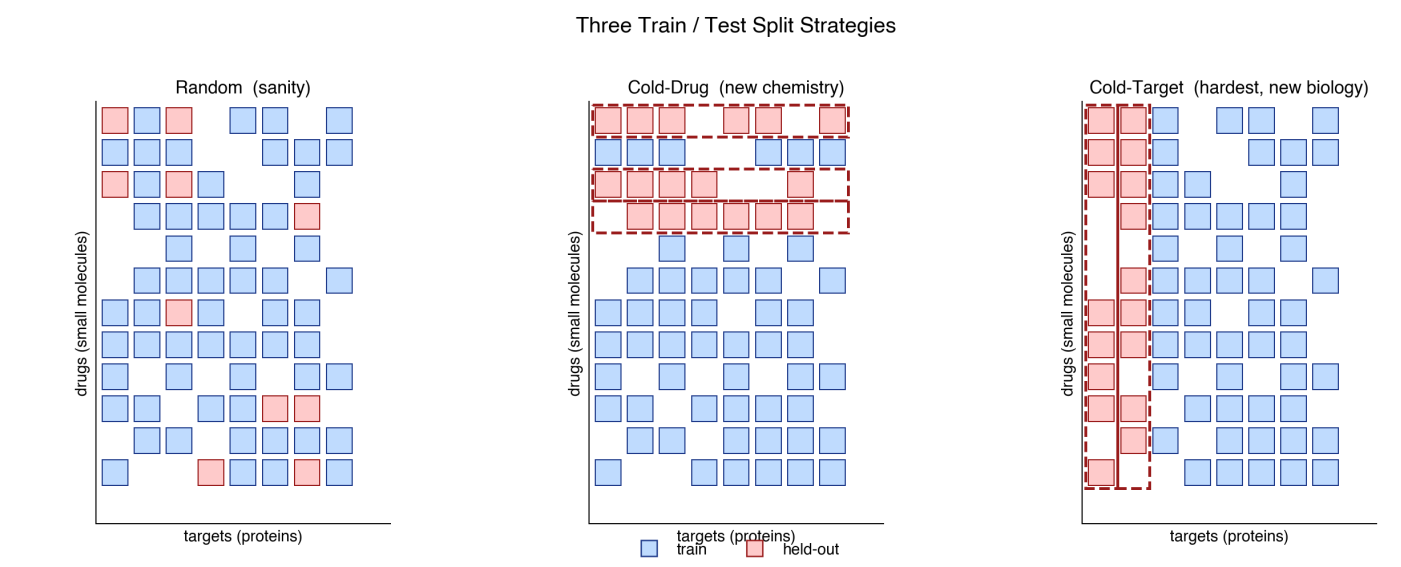


c) Data — BindingDB Ki (PDSP)



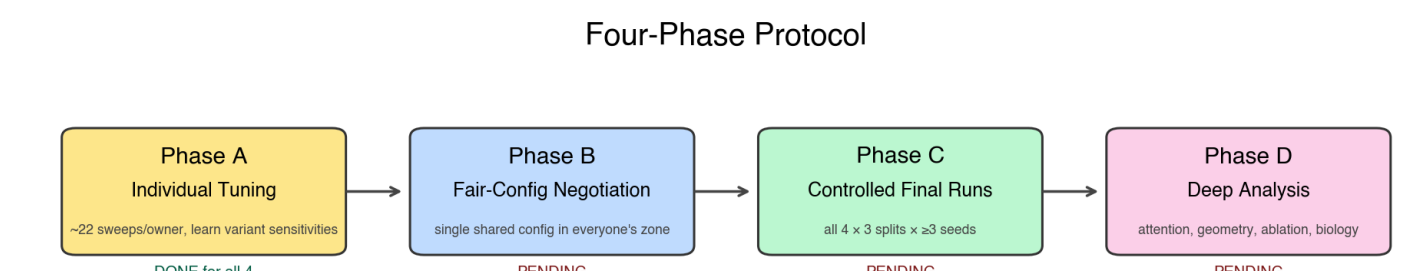
- 27,715 pairs.** $pKi = -\log_{10}(K_i/10^9) \in [3, 12]$. Caps: drug ≤ 100 tokens; protein ≤ 1200 res. Censored: 21% at $pKi = 5.0$; 17% drugs / 1% proteins truncated.

d) Three Splits — Random + Two Cold



Random 80/10/10 (sanity), **Cold-Drug** (new chemistry), **Cold-Target** (new biology, hardest). Cold splits are 40% harder on average — the inflation that random-split-only DTI papers hide.

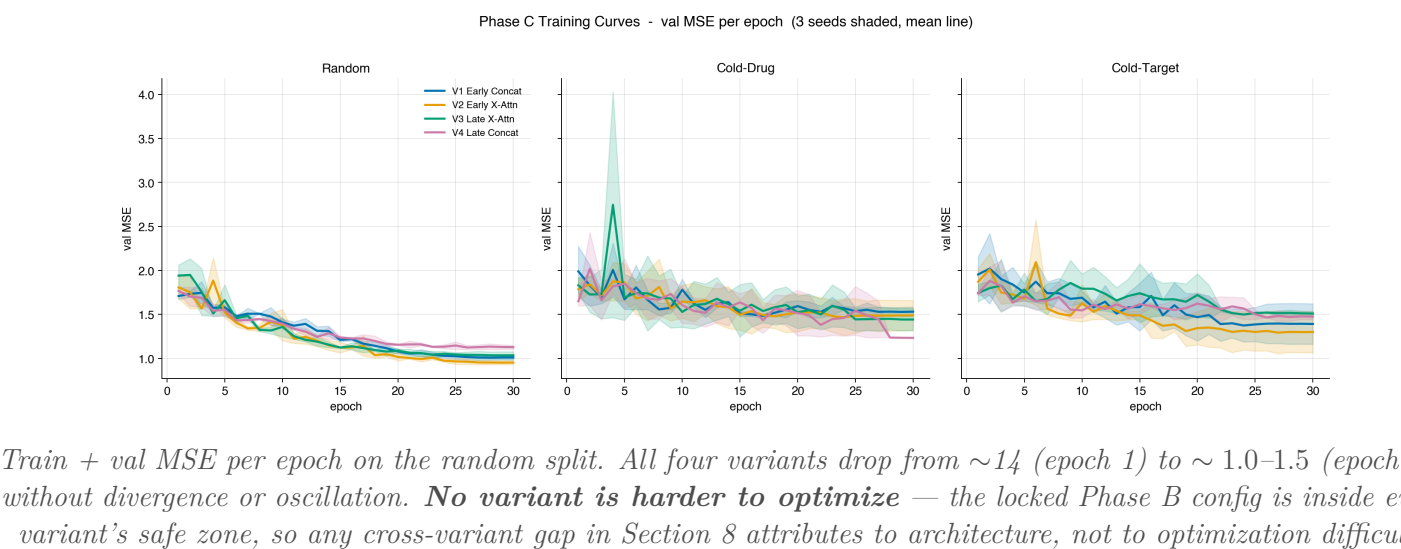
e) Four-Phase Protocol & Compute



A 88 sweeps $\rightarrow$ HP sensitivities. B Negotiate *fair config*. C 4 var $\times$ 3 splits $\times$ 3 seeds = **36 runs** at locked config. D Extract attention, predictions, hidden states, gradients, CKA.

Locked: $d=128$, $h=4$, $L=6/3+3$, $lr=3 \times 10^{-4}$, dropout 0.1, bs 64, 30 ep., AdamW + cosine + 500-step warmup, seeds 42/123/456. **Compute:** NYU HPC, $2 \times A100$, ~57 GPU-h (~4.7% of 1,200-h budget; ~7 kg CO₂-eq). **All four phases complete.**

f) Convergence Sanity — All Variants Train Cleanly



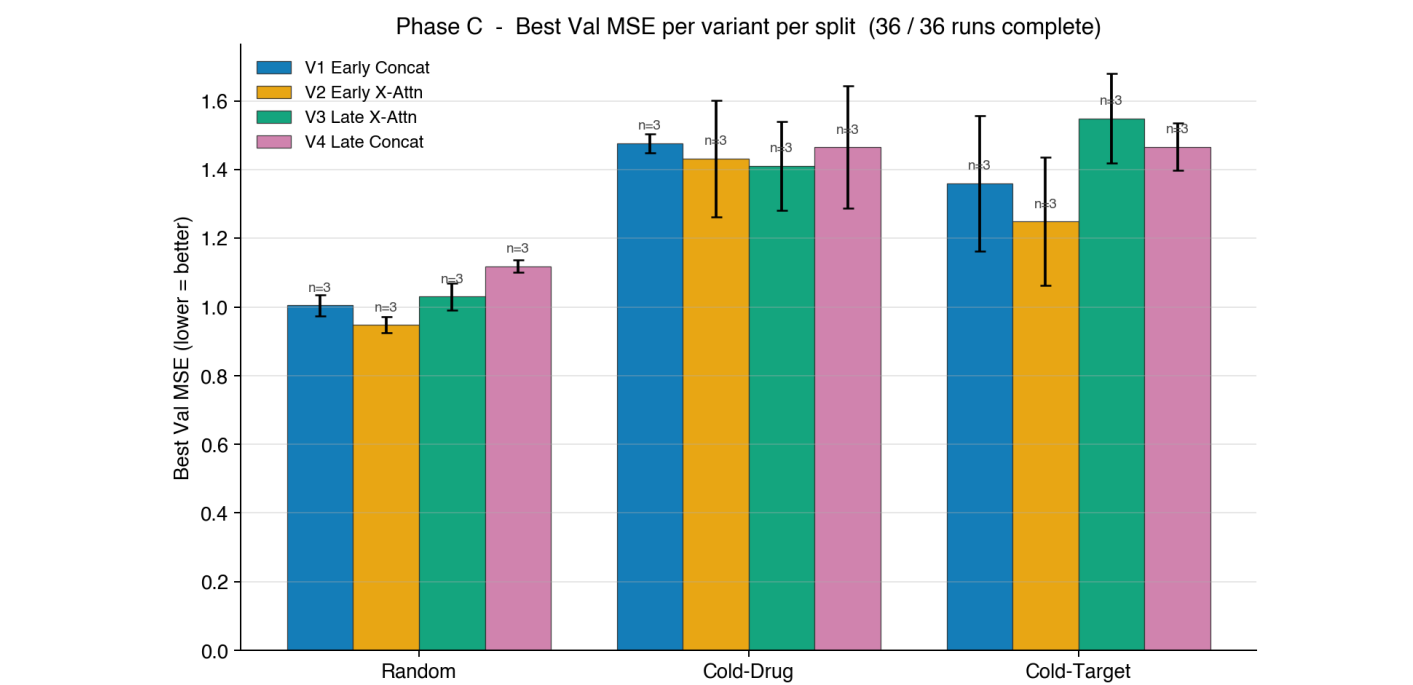
8. Results & Deep Analysis

a) Headline Numbers — Phase C, 36 Runs

Variant	Random $\downarrow$	Cold-Drug $\downarrow$	Cold-Tgt $\downarrow$
V1 Early Con	1.004 $\pm$ 0.031	1.476 $\pm$ 0.029	1.360 $\pm$ 0.197
V2 Early X-A	0.948 $\pm$ 0.023 $\star$	1.432 $\pm$ 0.170	1.248 $\pm$ 0.187 $\star$
V3 Late X-A	1.030 $\pm$ 0.039	1.410 $\pm$ 0.129 $\star$	1.549 $\pm$ 0.131
V4 Late Con	1.119 $\pm$ 0.018	1.465 $\pm$ 0.178	1.467 $\pm$ 0.069

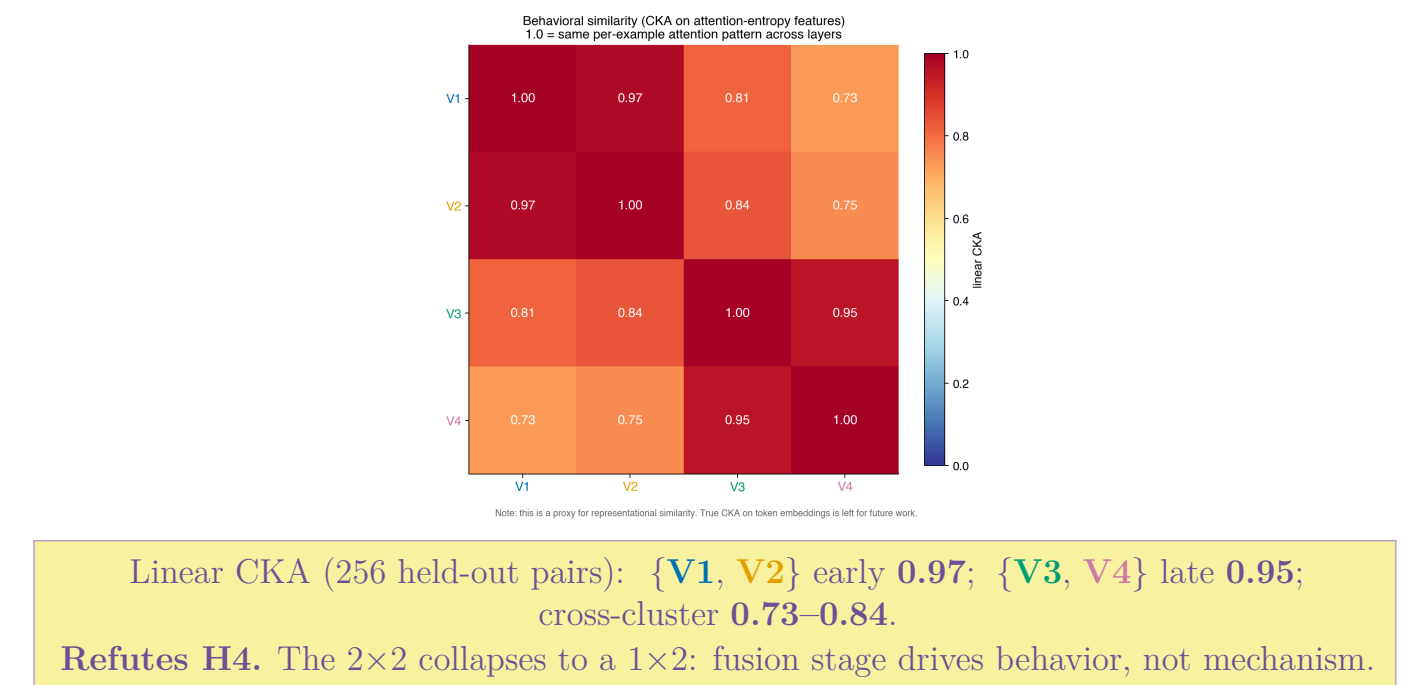
Mean $\pm$ std (3 seeds). $\star$ = best per column. **V2** wins 2 splits; **V3** wins cold-drug; **V4** (field default) never wins.

b) MSE per Split — All Four Variants

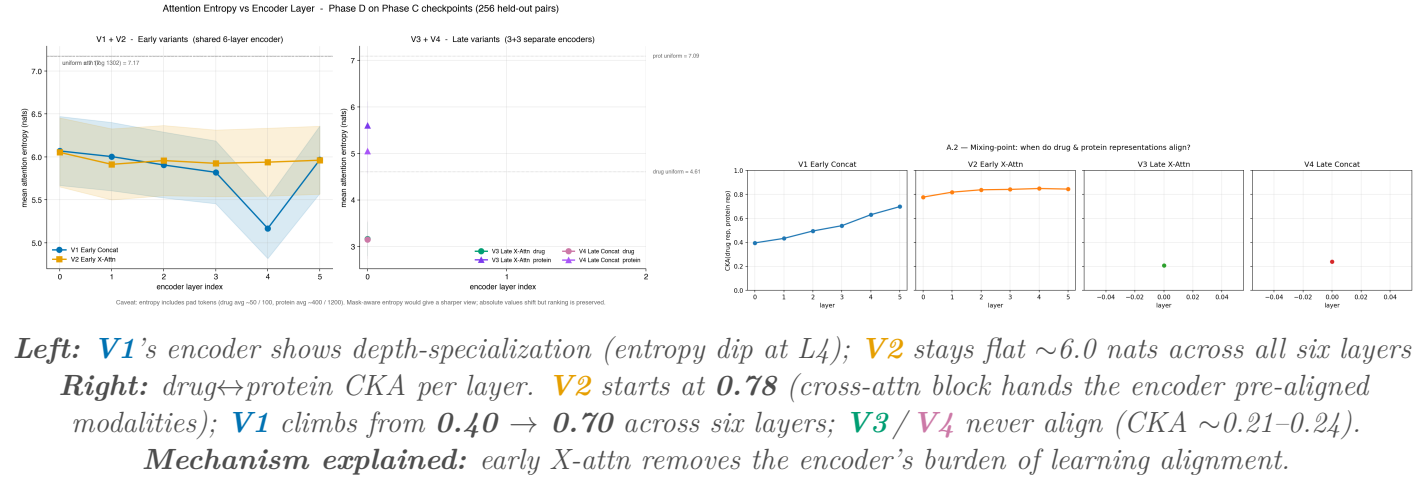


Bars = mean MSE across 3 seeds; whiskers = ± 1 std. Cross-attention (**V2**, **V3**) wins every split. Concatenation (**V1**, **V4**) wins none. Cold splits cost +40% MSE on average.

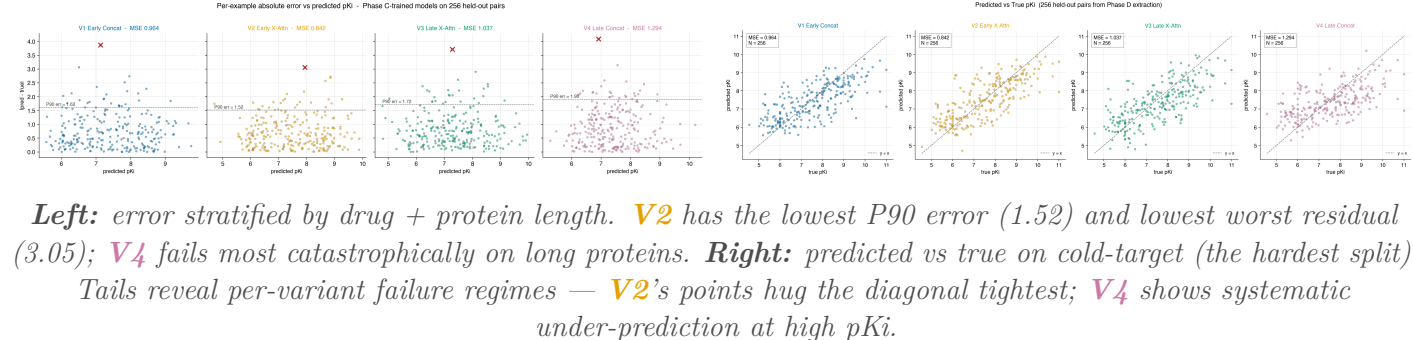
c) The CKA Finding — Stage $\succ$ Mechanism



d) Mechanism — Attention Entropy & Mixing-Point



e) Failure Modes — Where Each Variant Breaks



9. Findings & Conclusions

What we asked. Does the *stage* (before vs after encoding) and *mechanism* (concat vs X-attn) of cross-modal fusion change a transformer DTI model's accuracy, generalization, and learned representations?

What we found — five claims, each measured.

- No Pareto-optimal architecture.** Each split has a different winner. **V2** wins random + cold-target, **V3** wins cold-drug, **V4** (field default) wins zero. Architecture choice should depend on *deployment scenario*.
- Cross-attention > concatenation, on every split.** **V2 + V3** collectively win 3 of 3; **V1 + V4** win 0 of 3. The ~10% extra X-attn parameters earn their cost.
- Stage $\succ$ Mechanism (the deep finding).** CKA clusters by stage (within-cluster ≥ 0.95), not by mechanism (across-cluster 0.73–0.84). The 2×2 design collapses behaviorally to a 1×2 .
- Mechanism explained.** Mixing-point CKA shows **V2's** pre-encoder cross-attn block delivers drug+protein alignment of **0.78 at layer 0**, where **V1** takes 6 layers to climb from 0.40 to 0.70. **V2's** flat attention-entropy curve is the *symptom*; the upstream alignment is the *cause*.
- V1 is the parameter-efficient sweet spot; V4 is the most reliable.** **V1:** 1.27M params, single shared encoder, beats **V4** on every split. **V4:** lowest seed variance ($\sigma \leq 0.07$ on 2 of 3 splits) but highest mean MSE — usable when ranking-stability matters.

Why it matters. Switching from the field-default **V4** to early-fusion cross-attention (**V2**) yields a **15% relative MSE reduction on random** and **17% on cold-target** — the most realistic deployment scenario. The six-axis mechanistic dissection (information flow, geometry, attention, gradients, mixing-point, error modes) explains *why*, not just *what*.

10. Future Work

Limitations. Single-dataset training (BindingDB Ki, 21% censored at $pKi = 5.0$); sequence-only inputs (no 3D structure, no pre-trained encoders); modest scale ($d=128$, 1.2–1.4M params); causal head/layer ablations and binding-site recovery vs PDBbind program (PDSP) for the public dataset.

Future Work.

- Multi-modal drug input: SMILES + molecular graph (GNN) + 3D conformer.
- Pre-trained encoders (ChEMBERTa + ESM-2) — does pre-training neutralize the fusion-stage gap?
- Causal interventions on Phase C ckpts: head / layer ablation, representation swap (~3 GPU-h per variant).
- Binding-site recovery vs PDBbind: Precision@K on attention-as-attribution.
- Agentic drug optimization: best DTI model wrapped as a tool inside an LLM lead-optimization agent.

11. Acknowledgements

Acknowledgements. NYU HPC for cloud-burst GPU access (~57 GPU-h). Course staff of CSCI-2565 *Machine Learning*: Prof. Rajesh Ranganath (instructor); Nhi Nguyen, Riya Mahesh, Siddhant Mohan (TAs). BindingDB and the Psychoactive Drug Screening Program (PDSP) for the public dataset.

References. [1] Vaswani *et al.* *Attention is All You Need*, NeurIPS 2017. [2] Kornblith *et al.* *CKA*, ICML 2019. [3] Huang *et al.* *MolTrans*, Bioinformatics 2021. [4] Huang *et al.* *DeepPurpose*, Bioinformatics 2020. [5] Bai *et al.* *DrugBAN*, Nat. Mach. Intell. 2023. [6] Zhao *et al.* *HyperAttentionDTI*, Bioinformatics 2022. [7] Nguyen *et al.* *PerceiverCPI*, 2022. [8] Chen *et al.* *TransformerCPI*, Bioinformatics 2020. [9] Liu *et al.* *BindingDB 2023*, NAR 2023. [10] Davis *et al.*, Nat. Biotech. 2011. [11] Pahikkala *et al.*, Brief. Bioinform. 2015. [12] Mayr *et al.*, Chem. Sci. 2018. [13] Lin *et al.* *ESM-2*, Science 2023. [14] Chithrananda *et al.* *ChEMBERTa*, 2020. [15] Sundararajan *et al.* *Integrated Gradients*, ICML 2017.

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